



*Technical Manual
for
RC7121G1
Recirculating Chillers*

Total Thermal SolutionsTM



Serial Number: _____

Date of Purchase: _____

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Preface

Unpacking

Retain all cartons and packing material until the unit is operated and found to be in good condition. If the unit shows external or internal damage, or does not operate properly, contact the transportation company and file a damage claim. Under ICC regulations, this is your responsibility. The unit is fully tested with clean water. Although the system has been drained, some residual fluid may remain. This will not hinder your unit's performance.

Warranty Information

All Lytron chillers are warranted against defective parts and workmanship. Refer to end of manual for more details.

After Sale Support

Lytron is committed to customer service both during and after the sale. If you have any questions concerning the operation of your unit, contact our Customer Support Department. Before calling, please refer to the label on the rear of the unit to obtain the model and serial numbers.

Safety Precautions

General

This system is designed to provide fluid cooling only as specified in this manual. If used in manner other than as specified, the safety protection of the system may be impaired. Warnings are posted throughout the manual. These warnings are printed bold type. Read and follow these important instructions. Failure to observe these instructions or use other than specified may impair safety protection and can result in permanent damage to the unit, significant property damage, or personal injury or death.

Make sure you read and understand all instructions and safety precautions listed in this manual before operating your unit. If you have any questions concerning the operation of your unit or the information in this manual, please contact our Customer Support Department (see After-sale Support).

- **Never place the unit in a location where excessive heat, moisture, or corrosive materials are present.**
- **The unit construction provides protection against the risk of electrical shock by grounding appropriate metal parts. The protection may not function unless the power cord is connected to a properly grounded outlet. It is the user's responsibility to assure a proper ground connection is provided.**
- **Never connect the SUPPLY or RETURN fitting to your building water supply or any pressurized source.**
- **Performance of installation, operation, or maintenance procedures other than those described in this manual may result in a hazardous situation and may void the manufacturer's warranty.**
- **Transport the unit with care. Sudden jolts or drops can damage the plumbing lines.**
- **Observe all warning labels. Never remove warning labels.**
- **Never operate damaged or leaking equipment.**
- **Never operate the unit without water in the reservoir.**
- **Always turn off the unit and disconnect the power cord from the power source before performing any service or maintenance procedures, or before moving the unit.**
- **Never operate equipment with damaged power cords.**
- **Refer service and repairs to a qualified technician.**

Labels and Symbols



RETURN

This symbol is next to the connection where heated fluid is returning from the customer's machine.



SUPPLY

This symbol is next to the connection where chilled fluid is being supplied to the customer's machine.



This symbol indicates when the water level is low.



The Protective Earth (ground) Terminal



CAUTION 230 VAC

PRECAUTION 230 VAC

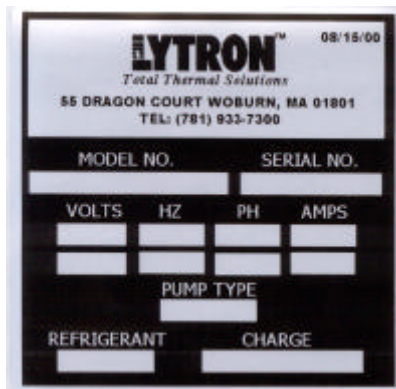
This symbol alerts maintenance personnel to consult the accompanying Installation and Operational Instructions Manual for an explanation of the Warning Symbol next to it. In this case, the High Voltage Warning on the terminal board cover brings about the need for this symbol.



This symbol certifies that we conform to the EMC Directive and all applicable directives at the time of marking.



This symbol is used to signify the power mode of the rocker style switch on the AC main line; Power On (I), Power Off (O).

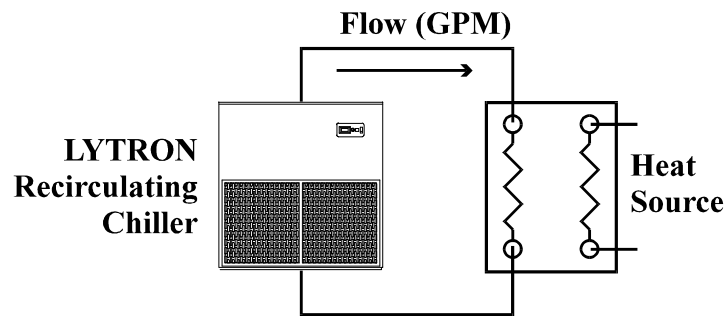


This label is placed on the machine to identify the model number, Volts, Amps, Hz, and Phase.

General Information

Chiller System Description

Turning the system switch to the “ON” position energizes the pump motor and system control circuit. The pump draws water from the internal reservoir, pumps it through the process fluid loop (removing heat), and returns it through the refrigeration system evaporator to the reservoir tank.



Process Fluid (flow control)

The RC models are designed to operate with continuous process fluid flow through a closed loop. This loop contains the system pump, temperature sensor, reservoir, internal & external plumbing lines and fittings, and the heat source. The external plumbing and the heat source are provided by the end user and are generally unique to the user's process and/or location. Some models are equipped with an internal flow bypass control. This control allows the user to maintain a particular flow rate through the process fluid loop.

Hot Gas Bypass

Starting and stopping can shorten a compressor's life much faster than steady running. For long life, our larger chillers use a refrigerant hot-gas bypass system. This provides capacity control by modulating refrigerant flow to the evaporator. This eliminates unnecessary on/off compressor cycling.

Specifications

Operating Environment	5°C - 40°C 80% RH
Unit Dimensions	width 21.4" x depth 27.8" x height 31.0"
Weight, Dry	270 lbs
Hose Connections	1/2" NPT
Input Power	208/230 V~ 60 Hz
	200 V~ 50 Hz
Current Draw	19.6 A @ 60 Hz
	19.4 A @ 50 Hz
Refrigerant Type	134A, 1 lbs 14 oz.
Thermal Capacity	4500 watts @ 20°C ambient air
Temperature Stability	+/- 0.5 °C
Pump Type	Positive Displacement

Installation

Quick Reference Start-Up Procedure

- 1) Record Model Number and Serial Number of the Chiller unit on the front cover of this manual.
- 2) Locate the Chiller unit as close to the heat load as possible leaving at least 18" around the case for air circulation and ventilation.
- 3) Remove all plastic caps covering fluid ports.
- 4) Attach process fluid lines to the inlet and outlet ports on the rear panel.
- 5) Connect Hubbell 2323 type power cord to the 230VAC, 60 Hz Hubbell 2325 power module.
- 6) Remove plate on top of the chiller unit and fill reservoir tank with clean water. Continue to add water until the system is filled and the reservoir remains at a constant level. Standard RC units may not be compatible with deionized water. Consult Lytron for deionized applications.
- 7) Turn on the chiller unit using the ON/OFF toggle switch located on the front panel.
- 8) Adjust the controller on the front display panel to the required output temperature. Press the "SET" button once to display the current set point value. Press the "UP" or "DOWN" button until the required output temperature appears on the LED display.
- 9) Check all fittings and hoses for fluid leaks. If a leak exists, turn chiller off and take necessary action to repair the leak.

Do not operate chiller until leak has been repaired.

Recording the Serial Number, Model Number, and Features

Record the serial number and the date purchased on the front cover of this manual. This information will be required when contacting Lytron for technical service. The serial number is located on the rear panel of the chiller.

Selecting Proper Location

Locate the chiller in an area with 18" minimum clearance around all sides of the unit for air circulation and ventilation. The chiller unit should be located as close as possible to the heat load to minimize pressure drop due to excessive line length. Cooling lines are best run at or near the same level as the cooling system, until reaching the equipment being cooling.

Plumbing to Process Fluid

The ports are located on the rear of the system and are labeled with the Supply  and Return  symbols. Connect the supply port to the input side of the heat load and the return port to the output of the heat load. The chiller process fluid supply and return ports are equipped with standard 1/4" MNPT fittings. When connecting fluid lines to the RC unit, use pipe sealant to ensure a leak-tight seal. Also, whenever possible use opaque lines to prevent algae buildup during prolonged non-operating periods.

All lines should be routed away from sources of heat and should be as short as possible. Lines which cannot be routed away from heat sources shall be protected with thermal insulating tape or other insulating material.

Never connect the SUPPLY or RETURN fitting to your building water supply or any pressurized source.

Flexible tubing, if used, should be of heavy wall or reinforced construction. All tubing should be rated to withstand 125 psig at $\pm 30^{\circ}$ C. Make sure all tubing connections are securely clamped. Avoid running tubing near radiators, hot water pipes, etc. If substantial lengths of tubing are necessary; insulation may be required to prevent loss of cooling capacity.

It is important to use the largest diameter tubing practical. Tubing should be straight, without bends and without diameter reductions. If substantial lengths of cooling lines are required, they should be pre-filled with cooling fluid before connecting them to the unit.

Fluid

Lytron recommends using clean water in the chiller unit. Pumps are susceptible to damage from abrasive materials in the process fluid.

- **Avoid using local water with high mineral content.**
- **Standard RC units may not be compatible with deionized water. Consult Lytron for deionized applications.**

Allowance should be made for the internal volume of cooling lines between the chiller and the equipment. The tank (reservoir) can be accessed by removing a panel on the top cover. Fill the tank and check level after operating for a short period of time. It may be necessary to add fluid if the level has dropped substantially.

NOTE: If the fluid is exposed to sunlight, add an algaecide to the fluid to control organic growth in lines.

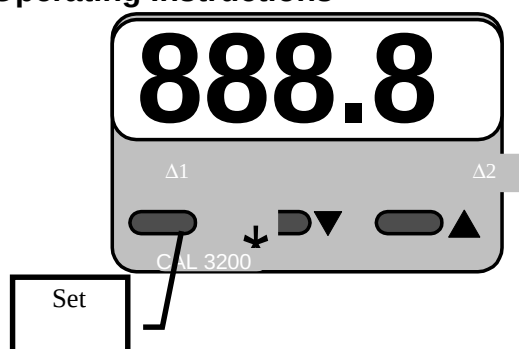
Electrical Requirements

Refer to the Specification section, and to the product label on the rear of the unit for the specific electrical requirements of your unit. The chiller power module is configured with a standard Hubbell 2325 inlet. A cord set with a Hubbell 2323 outlet plug and an inlet plug that is compatible to the local power grid and the power requirements of the chiller will be needed to safely operate the chiller.

- **It is the user's responsibility to ensure a proper ground connection is provided. Through the use of a properly grounded outlet.**

Operation

Digital Controller Operating Instructions



- 1) Prior to turning on the AC power switch, check the line cord and make sure that it is plugged in.
- 2) To operate the chiller, turn the AC power switch located on the front panel to the "ON" position. When the AC power switch is activated, the controller display will illuminate and display "8888" for 3 seconds and then display the temperature of the process cooling fluid. If the low level light illuminates, turn the chiller off and add more fluid to the reservoir.
- 3) Press and hold the "SET" button on the controller to display the current temperature set point. When the "SET" button is pressed and held, the controller display will alternate between "°C" and the current temperature setpoint.
- 4) To change the temperature set point, press and hold the "SET" button on the controller. While holding the "SET" button in, push the "UP" or "DOWN" button to adjust the temperature to the required set point value. The new temperature set point will be displayed on the controller when the "UP" ▲ and "DOWN" ▼ buttons are pressed.

Tune at Setpoint for Best Results

This procedure should be run under normal load conditions at desired setpoint temperature.

- 1) Turn on main power.
- 2) Press and hold "SET" (*) on controller and set desired temperature using up and down buttons.
- 3) After fluid reaches desired temperature press and hold both the up and down for 3 seconds, TUNE/OFF will flash on the display release.

- 4) On the controller press and hold "SET" (*) and using the up button select At.SP release.
- 5) Press and hold both the up and down buttons on the controller for 3 seconds. The display will continue to flash between TUNE, At.SP, and current temperature readings until all parameters are programmed into the controller. Note the temperature will fluctuate during this step while the controller is calculating the desired parameters. If TUNE/FAIL flashes on the controller repeat from step 3.
- 6) When the programming is complete only the current fluid temperature will be displayed.

If application changes or if controller is having difficulty maintaining the desired set point the above procedure should be repeated for new conditions.

System Maintenance / Service

Routine system maintenance should be very minimal once the RC7121G1 Chiller System has been initially installed, filled and is up and running.

Operating experience with each unique equipment installation will dictate reasonable frequency of system inspections and/or scheduled maintenance. Suggested guidelines are as follows:

Weekly Inspections

Noise Level

Any abnormal sound or substantial increase in noise level since the last weekly inspection may indicate an impending pump; fan or coolant blockage problem, which should be further, investigated.

Leakage

The weekly visual check does not require removal of the top cover. However, observation of coolant on the floor surface, coming out from under the system, calls for a further check for possible leaks.

Coolant Level

Any significant drop in the coolant level since the previous weekly check should be investigated further. If there is no visual system leak, then the loss may be due to leakage elsewhere in the equipment.

Pump

The pump requires no maintenance.

Pump Motor Lubrication

After 3 years of normal use or 1 year of heavy-duty service add oil annually. Use electric motor or SAE 10 oil.

Periodic Inspections

Algae

To restrict growth of algae in the reservoir, it is recommended that the reservoir cover be kept in place and that all circulation lines be opaque to ultraviolet. This will minimize the entrance of light, which is required for the growth of most common algae. Contact our Customer Support Department if algae becomes a problem.

Reservoir

Periodically inspect the fluid inside the reservoir. If cleaning is necessary, flush the reservoir with a cleaning fluid compatible with the circulating system and the cooling fluid. The cooling fluid should be replaced periodically. Build up of minerals and other contaminants occurs in clean water. It is important to flush, drain and refill every 3 months.

Trouble Shooting

Checklist

- Unit will not start

Check the line cord; make sure it is plugged in.

Check the voltage on the power source. Make sure it is within the rated voltage of the unit $\pm 10\%$

Check that the Power Switch is on and that the fuses have not blown.

- Inadequate capacity

Make sure the air intake and discharge are not impeded and the ambient temperature does not exceed $+40^{\circ}\text{C}$.

Make sure the air filter is free of dust and debris.

Service Assistance

If, after following these trouble-shooting steps, your unit fails to operate properly, contact our Service Department for assistance.

Warranty

Lytron agrees that the apparatus manufactured by it will be free from defects in materials and workmanship for the warranty period under normal use and service and when properly installed. The warranty period for recirculating chillers is two years from date of shipment of such apparatus to the original purchaser and one year from date of shipment of such apparatus to the original purchaser for all other products Lytron sells. Lytron's obligation under this agreement is limited solely to repair or replacement, at its option, at its factories, of any part or parts thereof, returned to Lytron with transportation charges prepaid, which examination shall disclose to Lytron's satisfaction to have been defective. THE FOREGOING EXPRESS WARRANTY IS IN LIEU OF ALL OTHER WARRANTIES, EXPRESSED OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OR MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE. LYTRON'S OBLIGATION UNDER THIS WARRANTY IS STRICTLY AND EXCLUSIVELY LIMITED TO THE REPAIR OR REPLACEMENT OF DEFECTIVE COMPONENT PARTS AND LYTRON DOES NOT ASSUME OR AUTHORIZE ANYONE TO ASSUME FOR IT ANY OTHER OBLIGATION. LYTRON ASSUMES NO RESPONSIBILITY FOR INCIDENTAL, CONSEQUENTIAL, OR OTHER DAMAGES INCLUDING, BUT NOT LIMITED TO LOSS OR DAMAGE TO PROPERTY, LOSS OF PROFITS OR REVENUE, LOSS OF THE UNIT, LOSS OF TIME, OR INCONVENIENCE. Lytron's liability does not include any labor charges for replacement of parts, adjustments, repairs, or any other work done outside its factories and its liability does not include any resulting damage to persons, property, equipment, goods or merchandise arising out of any defect in or failure of its apparatus. Lytron's obligation to repair or replace shall not apply to any apparatus which shall have been repaired or altered outside of its factory in any way, or which has been subject to negligence, to misuse, or to pressures in excess of stated limits. On parts not of Lytron's manufacture, such as motors, controls, etc., Lytron extends only those warranties given to Lytron, Inc. to the extent Lytron can do so. Lytron's agreement hereunder runs only to the immediate purchaser from Lytron, Inc. and does not extend, expressly or by implication, to any other person.